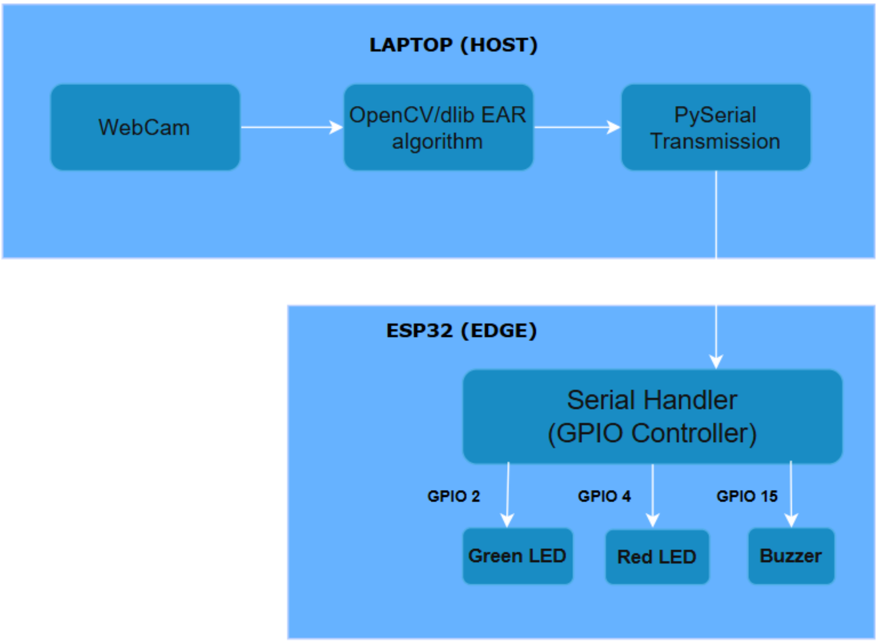
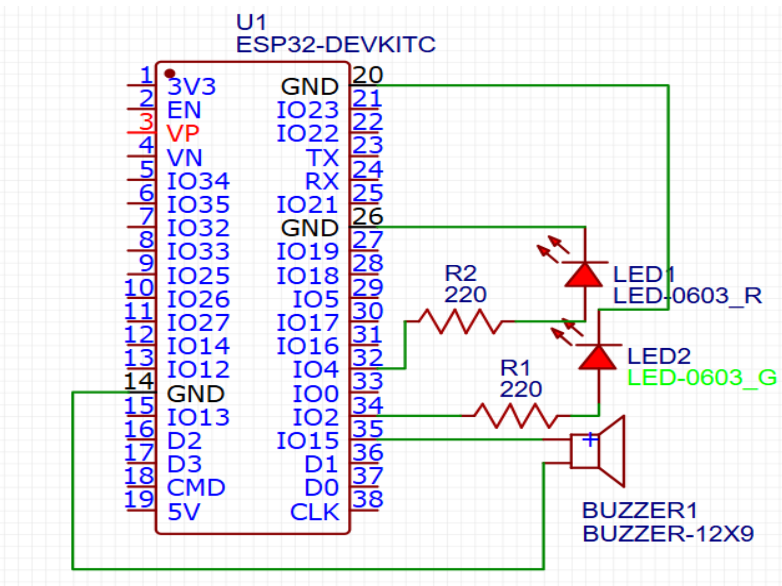
1.Problem Statement: Driver fatigue and drowsiness are leading causes of road accidents globally. A real-time, non-invasive system is required to monitor driver alertness and provide instant hardware alerts to prevent catastrophic events.

2. Scope of the solution: This project implements a hybrid Edge-IoT prototype. Image acquisition and computer vision processing (EAR calculation) are executed on a host machine, while real-time peripheral alerting mechanisms are offloaded to an ESP32 microcontroller via UART communication.

3. Block Diagram:



4. Circuit Diagram:



5. Required Components (IDE, Software, Hardware):

IDE: VS Code, Arduino IDE

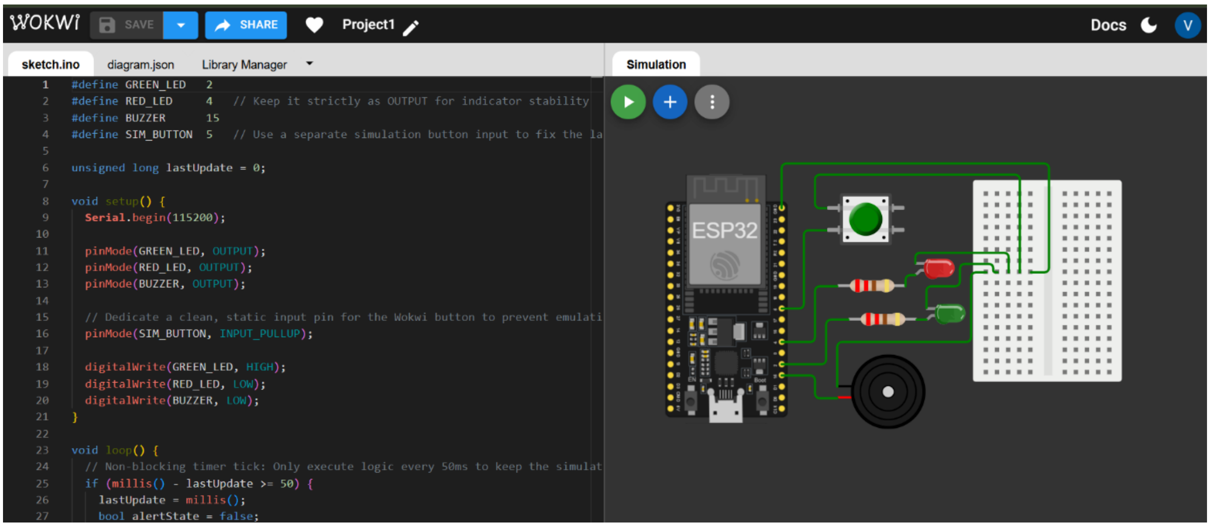
Software: Wokwi, EasyEDA

BILL OF MATERIALS

S. No	Name of Component	Quantity	Cost
1	ESP32 Dev Kit C	1	₹ 670/-
2	Buzzer	1	₹ 5/-
3	LEDs	2	₹ 2/-
4	Resistors (220 Ω)	2	₹ 2/-
5	USB Cable	1	₹ 50/-
6	Breadboard	1	₹ 80/-
7	Jumper Wires (Male to Female)	7	₹ 7/-

Total Cost: ₹ 816/-

6. Simulated Circuit:



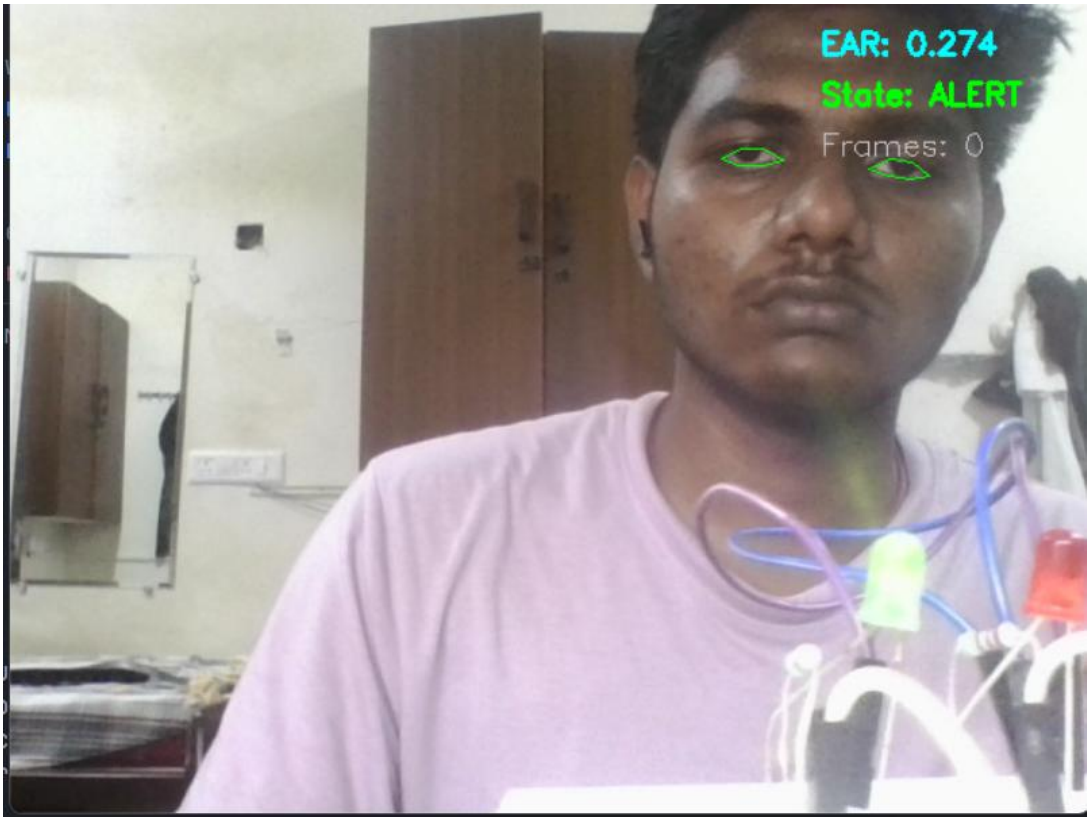
*Used push button in simulation as Wokwi doesn't support camera.

7. Code for the solution:

https://github.com/vaasanthpendem/Drowsiness-Detection-System.git

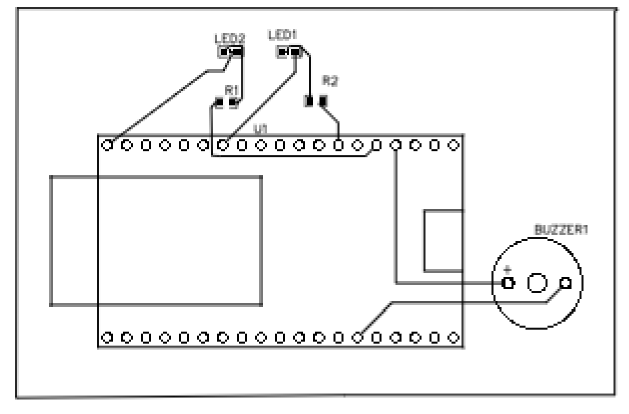
8. Results of the simulation:

State of being ‘**alert’**, **blink of green LED**



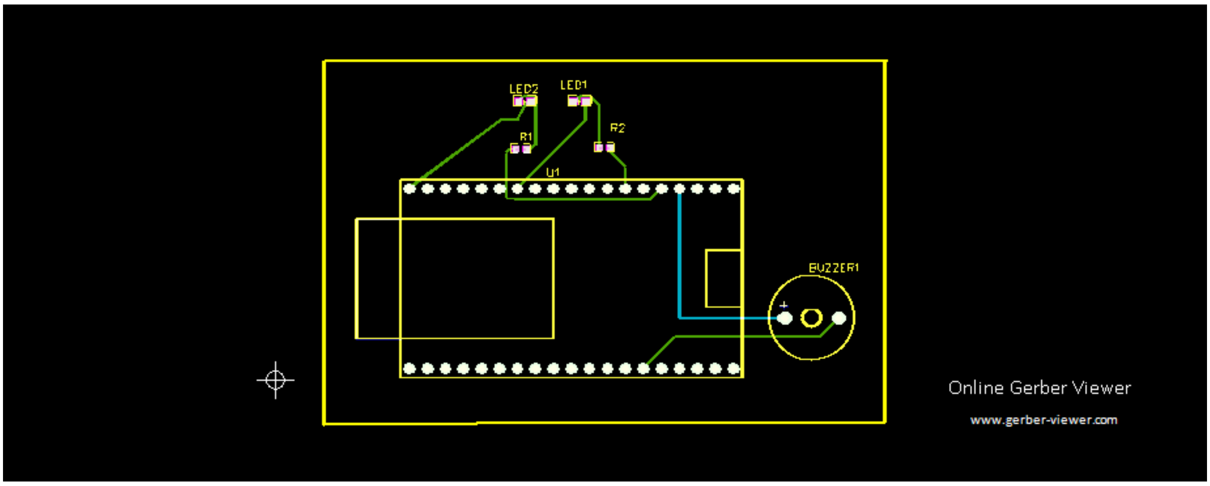
Drowsiness detected, blink of **red LED** with buzzer **‘beep’** alarm

9. Layout of the system:



10. Gerber Files:

Uploaded in **GitHub repo** as .zip format.
https://github.com/vaasanthpendem/Drowsiness-Detection-System.git



11. Video demo

https://drive.google.com/file/d/1n0hNS0R_YoxeVFXQ7voELo-CGKAu_B7I/view?usp=drive_link

12. References

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